## 7.6 Two Position Vectors and Time-Lambert Problem

- (General) Lambert Problem
  - Known: Two Position Vectors and Time of Flight between each position
  - Unknown: Orbit between the end points

  - Technique for Initial Orbit Determination
  - Transfer Problem (Rendezvous and Intercept)
  - Correct way to analyze complex maneuvers introduced in Chapter 6.

- Lambert Original Formulation
  - Geometrically interprets the problem
  - Provides a way to determine the Minimum-Energy Transfer between the two known endpoints
  - Requires to know the Angle through which the transfer takes place (Fig 7-8)

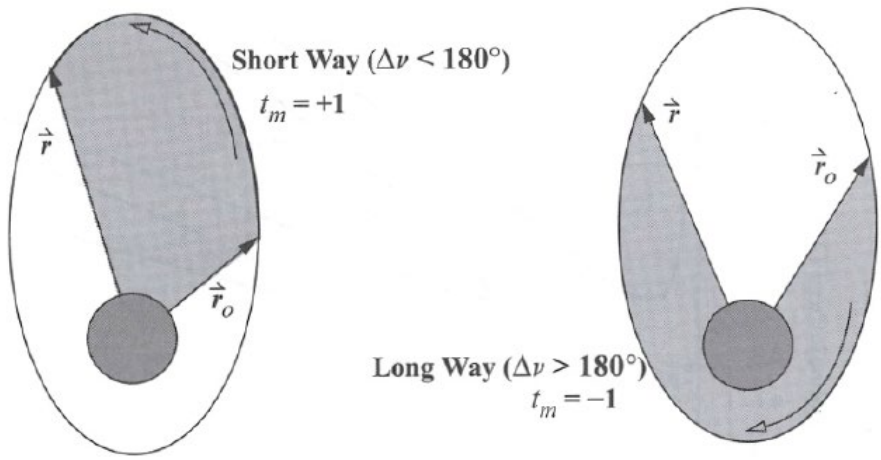



**Figure 7-8. Transfer Methods, $t_m$, for the Lambert Problem.** Traveling between the two specified points can take the long way or the short way. For the long way, the change in true anomaly exceeds 180°.

In determining the transfer angle of angle between two position observations in the original orbit determination problem, we assume a direction

$$\cos(\Delta\nu) = \frac{\vec{r}_0 \cdot \vec{r}}{r_0 r}$$

$$\sin(\Delta\nu) = t_m \sqrt{1 - \cos^2(\Delta\nu)}$$

In these relations, recognize that $t_m$=+1 for short-way transfer and $t_m$=-1 for long-way transfer.

- The core idea of Lambert Problem use the following relation to locate the other focus and determine the orbit's semimajor axis (Fig. 7-10):

$$r + r' = 2a \Rightarrow r' = 2a - r$$
$$r_0 + r_0' = 2a \Rightarrow r_0' = 2a - r_0$$

  - Draw circles about each position with radii $2a - r$ and $2a - r_0$
  - Intersections of corresponding circles mark the possible locations of the other focus
  - Fig 7-10 shows the foci for one semimajor axis, along with a corresponding orbit

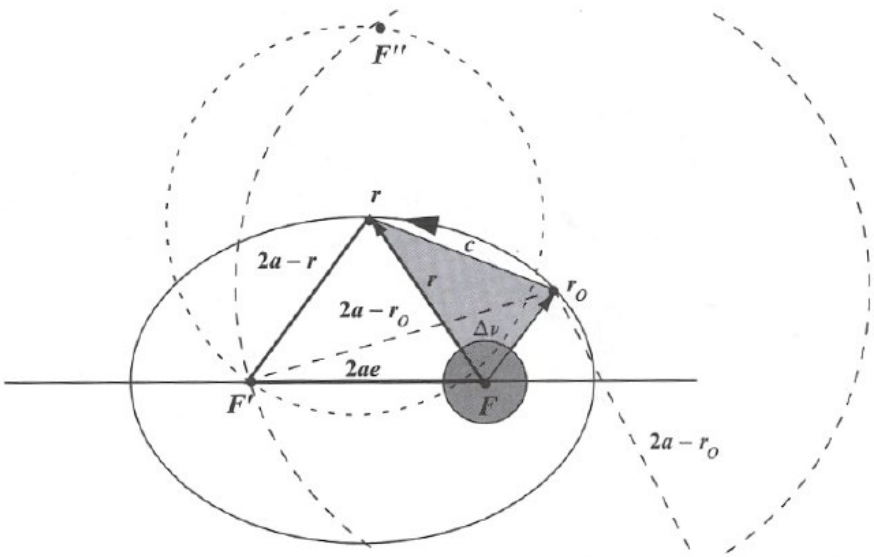



**Figure 7-10. Geometry for the Lambert Problem (I).** This figure shows how we locate the secondary focus—the intersection of the dashed circles. The chord length, $c$, is the shortest distance between the two position vectors. The sum of the distances from the foci to any point, $r$ or $r_o$, is equal to twice the semimajor axis.

### 7.6.1 Lambert–Minimum Energy

- Lambert-Minimum Energy Orbit
  - Fig 7-11 shows many circles representing different values of $a$
  - Minimum Energy Orbit is obtained when the chord length $c$ between two position vectors equals the sum of the two radii
    - Note that the Energy $E = -\dfrac{\mu}{2a}$ becomes minimum when $a$ is the smallest

The **semiperimeter *s*** is defined as half the sum of three sides of the triangle created by two position vectors and the chord:

$$s = \frac{r_0 + r + c}{2}$$

The length between two foci is given from Fig 7-10.

$$FF' = 2ae$$

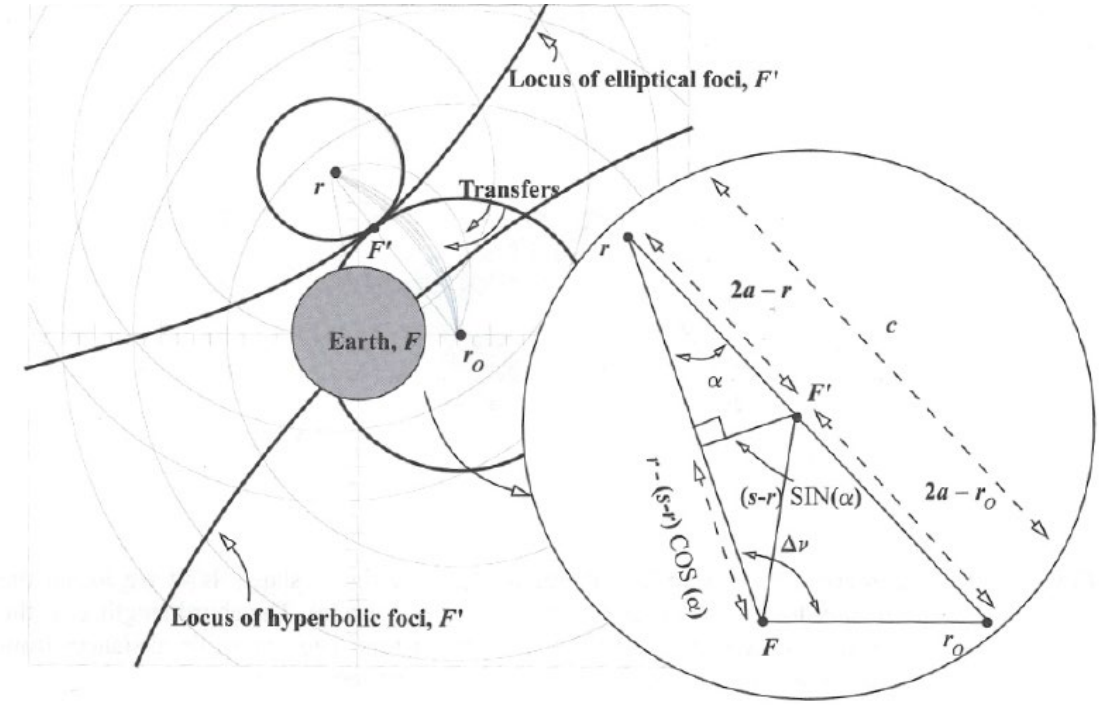



**Figure 7-11. Geometry for the Lambert Problem (II).** Solving Lambert's problem relies on many geometrical quantities. Be sure to allow for the Earth when viewing representations like this. I've shown transfers between a satellite at 9567.2 km (1.5 ER) and 15,307.5 km (2.4 ER) from the Earth's center. We can use the inset figure to find the transfer semimajor axis. The concentric circles are drawn for elliptical values of *a*. When the circles (of the same *a*) touch, half the sum of their radii equals *a* of the transfer, and the intersection is the location of the second focus, *F'*.

$$(2a_{\min} - r) + (2a_{\min} - r_0) = c$$

$$\Rightarrow 4a_{\min} = r_0 + r + c$$

$$\Rightarrow a_{\min} = \frac{r_0 + r + c}{4} \qquad \left( \Leftarrow \; s = \frac{r_0 + r + c}{2} \right)$$

$$\Rightarrow 2a_{\min} = s$$

$$\Rightarrow 2a_{\min} - r = s - r, \quad 2a_{\min} - r_0 = s - r_0$$

$$\begin{aligned}
4a_{\min}^2 e_{\min}^2 = (FF'_{\min})^2 &= [(s-r)\sin\alpha]^2 + [r - (s-r)\cos\alpha]^2 \\
&= (s-r)^2 \sin^2\alpha + (s-r)^2 \cos^2\alpha + r^2 - 2r(s-r)\cos\alpha \\
&= (s-r)^2 + r^2 - 2r(s-r)\cos\alpha = s^2 - 2rs + r^2 + r^2 - 2r(s-r)\cos\alpha \\
&= s^2 - 2r(s-r) - 2r(s-r)\cos\alpha = s^2 - 2r(s-r)(1+\cos\alpha)
\end{aligned}$$

(7-38)

$$(1+\cos\alpha) = ??$$

*Consider*

$$2s(s-r_0)-rc = 2\frac{r_0+r+c}{2}\left(\frac{r_0+r+c}{2}-r_0\right)-rc \qquad \left(\Leftarrow s=\frac{r_0+r+c}{2}\right)$$

$$=\frac{1}{2}(-r_0^2+r_0r+r_0c-r_0r+r^2+rc-r_0c+rc+c^2)-rc$$

$$=\frac{-r_0^2+r^2+c^2}{2} \qquad \left(\Leftarrow \frac{-r_0^2+r^2+c^2}{2}=rc\cos\alpha \qquad \Leftarrow \cos\alpha=\frac{r^2+c^2-r_0^2}{2rc}\right)$$

$$=rc\cos\alpha$$

$$\Rightarrow \cos\alpha=\frac{2s(s-r_0)}{rc}-1$$

$$\Rightarrow 1+\cos\alpha=\frac{2s(s-r_0)}{rc} \tag{1}$$

Introducing (1) into (7-38) results in

$$4a_{\min}^2e_{\min}^2 = s^2-2r(s-r)\frac{2s(s-r_0)}{rc}$$

$$\Rightarrow 4a_{\min}^2e_{\min}^2 = s^2-\frac{4s}{c}(s-r_0)(s-r) \tag{2}$$

*Consider* $\quad 4a^2e^2 = 4a\left(ae^2\right)$

$$\left(\Uparrow ae^2=a-p \quad \Leftarrow p=a(1-e^2)\right)$$

$$=4a(a-p)$$

$$\left(\Uparrow \ a=a_{\min}=\frac{s}{2}, e=e_{\min}, p=p_{\min}\right)$$

$$\Rightarrow 4a_{\min}^2e_{\min}^2 = 4\frac{s}{2}\left(\frac{s}{2}-p_{\min}\right)$$

$$=s^2-2sp_{\min} \tag{3}$$

Introducing (3) into the left hand side of (2) leads to

$$s^2-2sp_{\min} = s^2-\frac{4s}{c}(s-r_0)(s-r)$$

$$\Rightarrow p_{\min} = \frac{2}{c}(s - r_0)(s - r)$$

$$= \frac{2}{c}\left(\frac{r + r_0 + c}{2} - r_0\right)\left(\frac{r + r_0 + c}{2} - r\right)$$

$$= \frac{1}{2c}[2rr_0 + c^2 - r^2 - r_0^2]$$

$$\left(\Uparrow\ c^2 = r^2 + r_0^2 - 2rr_0 \cos\Delta\nu\right)$$

$$\Rightarrow p_{\min} = \frac{rr_0}{c}(1 - \cos\Delta\nu) \tag{4}$$

$$e = \sqrt{1 - \frac{p}{a}} \tag{2-78}$$

$$\Rightarrow e_{\min} = \sqrt{1 - \frac{p_{\min}}{a_{\min}}} = \sqrt{1 - \frac{2p_{\min}}{s}} \tag{5}$$

$$\vec{r} = f\vec{r}_0 + g\vec{v}_0$$

$$\Rightarrow\ \vec{v}_0 = \frac{1}{g}(\vec{r} - f\vec{r}_0) \tag{2-57}$$

$$\Rightarrow\ \vec{v} = \dot{f}\vec{r}_0 + \dot{g}\vec{v}_0$$

$$f = 1 - \frac{r}{p}[1 - \cos\Delta\nu], \qquad g = \frac{rr_0 \sin\Delta\nu}{\sqrt{\mu p}}, \tag{2-63}$$

$$\dot{f} = \sqrt{\frac{\mu}{p}} \tan\left(\frac{\Delta\nu}{2}\right)\left(\frac{1 - \cos\Delta\nu}{p} - \frac{1}{r} - \frac{1}{r_o}\right),\ \dot{g} = 1 - \left(\frac{r_o}{p}\right)(1 - \cos\Delta\nu)$$

$$\Rightarrow \vec{v}_0 = \frac{1}{g}(\vec{r} - f\vec{r}_0) = \frac{\sqrt{\mu p_{\min}}}{rr_0 \sin\Delta\nu}[\vec{r} - \{1 - \frac{r}{p_{\min}}(1 - \cos\Delta\nu)\}\vec{r}_0] \tag{6}$$

Then, we finally know $(\vec{r}_0, \vec{v}_0)$ *and* $(\vec{r}, \vec{v})$ which can be used for initial orbit determination or orbital maneuvers.

- Summary: Lambert–Minimum Energy Problem
  - Given $\left(\vec{r}_0, \vec{r}\right)$
  - Calculate $a_{\min}, e_{\min}, p_{\min}, t_{\min}, \vec{v}_0, \vec{v} = \dot{f}\vec{r}_0 + \dot{g}\vec{v}_0$

- Measure $\Delta\nu$
- Calculate the Chord Length $c$ and Semiperimeter $s$
- Use (4), (5), (3), (7) and (6) to find $p_{\min}, e_{\min}, a_{\min}, t_{\min}$ and $\vec{v}_0$, respectively
- Find $\vec{v} = \dot{f}\vec{r}_0 + \dot{g}\vec{v}_0$

Flight Time

$$t_{\min} = \sqrt{\frac{a_{\min}^3}{\mu}}\left[(\alpha_e - \sin\alpha_e) - (\beta_e - \sin\beta_e)\right] \text{ short way} \tag{7}$$

where $\sin\frac{\alpha_e}{2} = \sqrt{\frac{s}{2a}}$, $\sin\frac{\beta_e}{2} = \sqrt{\frac{s-c}{2a}}$

proof: (Marshall H. Kaplan, *Modern Spacecraft Dynamics & Control*, 1976, John Wiley & Sons, pp. 319-322.)

### 7.4.2 Lambert's Time-of-Flight Theorem

A very important property of the flight from $P$ to $Q$ is the time. In fact, the flight time must be related to the type of path taken. Lambert discovered that this time depends only on the value of $a$ for a given situation. He postulated that traverse time is a function of major axis, the sum of distances between center of force and end points, and the length of the chord between these end points, which in functional form is

$$t = t(a, r_1 + r_2, c)$$

Thus, time is independent of $p$ and $e$. Lagrange later proved this theorem analytically, and its validity is assumed here. Analytical formulation of flight time is possible by noting that since $P$ and $Q$ are fixed, the time of flight is also fixed as long as $r_1 + r_2 =$ constant and $a$ is constant. This implies that $F$ and $F^*$ may be moved around as long as $(r_1 + r_2)$ and $a$ remain unchanged. Thus, the locus of $F$ is $r_1 + r_2 =$ constant. (i.e., an ellipse with foci at $P$ and $Q$). This is illustrated in Figure 7.27. The major axis of this ellipse is $(r_1 + r_2)$. Now the locus of $F^*$ is $PF^* + QF^* = 4a - (r_1 + r_2)$ which is a constant. This locus is also an ellipse with foci at $P$ and $Q$ and major axis $[4a - (r_1 + r_2)]$. Figure 7.27 shows this locus and possible corresponding pairs of $F$ and $F^*$. A limiting case occurs when $F$ and $F^*$ are at extremities of their respective ellipses as represented by $F_2$, $F_2^*$. This corresponds to rectilinear motion between $P$ and $Q$. In general, the relative positions of $F$ and $F^*$ on their respective ellipses must satisfy

$$QF^* + QF = PF^* + PF = 2a$$

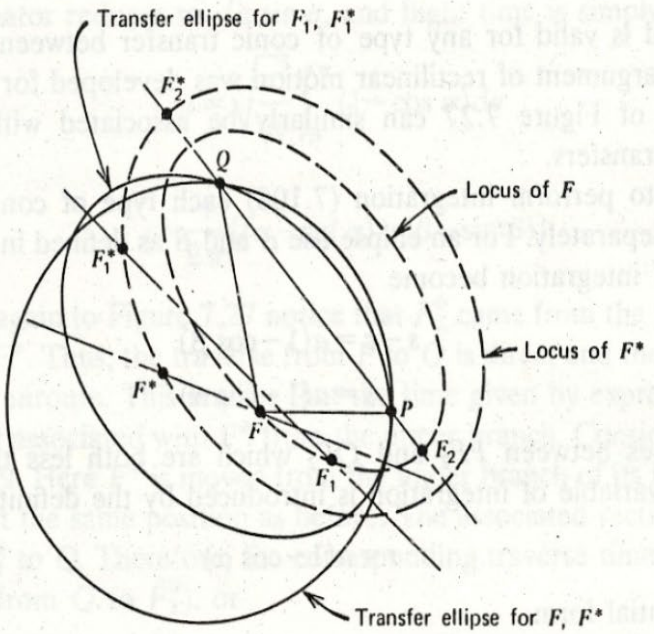



**FIGURE 7.27** Geometry related to Lambert's theorem.

The preceding geometrical innovation permits a straightforward calculation of flight time. For rectilinear motion velocity magnitude is simply

$$v = \frac{dr}{dt}$$

and energy gives

$$v^2 = \mu\left(\frac{2}{r} - \frac{1}{a}\right)$$

Thus, time is extracted from

$$dt = \frac{1}{\sqrt{\mu}} \frac{r\,dr}{\sqrt{2r - r^2/a}} \tag{7.107}$$

The geometry of this situation dictates

$$QF_2 + PF_2 = r_1 + r_2$$

$$QF_2 - PF_2 = c$$

Take the difference and the sum of these equations to extract

$$PF_2 = s - c$$

$$QF_2 = s$$

which are the end points of the time integration of expression (7.107),

$$t = \frac{1}{\sqrt{\mu}} \int_{s-c}^{s} \frac{r\,dr}{\sqrt{2r - r^2/a}} \tag{7.108}$$

This integral is valid for any type of conic transfer between $P$ and $Q$, even though the argument of rectilinear motion was developed for an elliptic path. $F_2$ and $F_2^*$ of Figure 7.27 can similarly be associated with parabolic and hyperbolic transfers.

In order to perform integration (7.108) each type of conic path must be considered separately. For an ellipse use $\alpha$ and $\beta$ as defined in set (7.99). Then the limits of integration become

$$\text{Define} \quad \sin\frac{\alpha}{2} = \sqrt{\frac{s}{2a}}, \quad \sin\frac{\beta}{2} = \sqrt{\frac{s-c}{2a}}$$

$$\sin\frac{\alpha}{2} = \sqrt{\frac{s}{2a}}, \quad \sin^2\frac{\alpha}{2} = \frac{s}{2a}, \quad s = 2a\sin^2\frac{\alpha}{2} = a(1-\cos\alpha)$$

$$s - c = a(1-\cos\beta)$$

$$s = a(1-\cos\alpha)$$

Since $r$ varies between $PF_2$ and $QF_2$ which are both less than $2a$, a more convenient variable of integration is introduced by the definition

$$r = a(1-\cos\sigma)$$

with differential form

$$dr = a\sin\sigma\, d\sigma$$

The limits of integration take on simple forms,

$$\text{at} \quad r = s - c, \sigma = \beta$$

$$\text{at} \quad r = s, \quad \sigma = \alpha$$

which brings expression (7.108) to

$$t = \frac{1}{\sqrt{\mu}}\int_\beta^\alpha \frac{a(1-\cos\sigma)(a\sin\sigma\, d\sigma)}{\left[2a(1-\cos\sigma) - \dfrac{a^2(1-\cos\sigma)^2}{a}\right]^{1/2}}$$

The denominator reduces to $\sqrt{a}\sin\sigma$, and flight time is simply

$$t = \sqrt{\frac{a^3}{\mu}}\int_\beta^\alpha (1-\cos\sigma)\, d\sigma \tag{7.109}$$

which yields

$$t = \frac{\tau}{2\pi}[(\alpha - \sin\alpha) - (\beta - \sin\beta)] \tag{7.110}$$

$\tau$ is the orbital period.

### 7.6.2 Lambert-Gauss Solution ($\vec{r}_0, \vec{r}, \Delta t \longrightarrow \vec{v}_0, \vec{v}$)

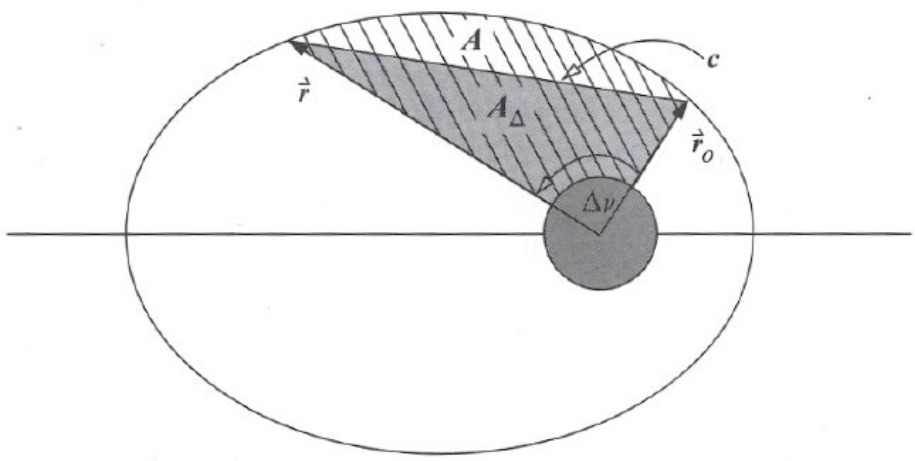
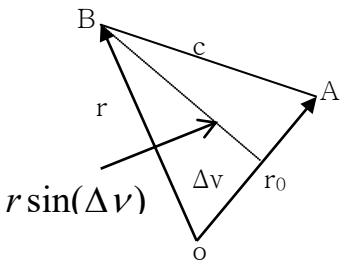



**Figure 7-12. Area Swept Out by a Satellite During Transfer.** Gauss's method relies on the area swept out by the satellite during its orbit. Two regions are useful—the triangular shaded region, $A_\Delta$, and the total area, $A$, swept out by the satellite.

- Orbit Determination from Two Position Vectors




The area of triangle OAB $(A_\Delta)$

$$A_\Delta = \frac{1}{2} r_0 r \sin(\Delta\nu)$$

The area of the sector of the ellipse by $\vec{r}_0$ and $\vec{r}$ is obtained from the Kepler's 2nd law:

$$\frac{dA}{dt} = \frac{h}{2} = \frac{\sqrt{\mu p}}{2}$$

$$\Rightarrow A = \frac{\sqrt{\mu p}\Delta t}{2}$$

Define the ratio $y$ as

$$y \equiv \frac{A}{A_\Delta} = \frac{\sqrt{\mu p} \cdot \Delta t}{r_0 r \sin(\Delta \nu)} \tag{1}$$

$$r_0 = \frac{p}{1 + e\cos\nu_0}, \quad r = \frac{p}{1 + e\cos\nu}$$

$$\Rightarrow \frac{p}{r_0} = 1 + e\cos\nu_0, \quad \frac{p}{r} = 1 + e\cos\nu$$

$$\Rightarrow p\left(\frac{1}{r_0} + \frac{1}{r}\right) = 2 + e(\cos\nu_0 + \cos\nu)$$

$$\left(\Uparrow \cos\nu_0 + \cos\nu = 2\cos\left(\frac{\nu_0 + \nu}{2}\right)\cos\left(\frac{\nu - \nu_0}{2}\right)\right)$$

$$\Rightarrow p\left(\frac{1}{r_0} + \frac{1}{r}\right) = 2 + 2e\cos\left(\frac{\nu_0 + \nu}{2}\right)\cos\left(\frac{\nu - \nu_0}{2}\right) \tag{2}$$

Note that $e\cos\left(\frac{\nu_0 + \nu}{2}\right)$ is unknown

The following steps are used to replace this factor with known quantities, i.e., Equation (3)

$$\begin{cases} r = a(1 - e\cos E) \\ r\cos\nu = a\cos E - ae = a(\cos E - e) \end{cases}$$

$$\Rightarrow \begin{cases} r(1 + \cos\nu) = r + r\cos\nu = a(1 - e)(1 + \cos E) \\ r(1 - \cos\nu) = r - r\cos\nu = a(1 + e)(1 - \cos E) \end{cases}$$

$$\cos^2\frac{\nu}{2} = \frac{1 + \cos\nu}{2}$$

$$\Rightarrow \sqrt{r}\cos\frac{\nu}{2} = \pm\sqrt{\frac{r(1 + \cos\nu)}{2}}$$

$$= \pm\sqrt{\frac{1}{2}a(1 - e)(1 + \cos E)}$$

$$\left(\Uparrow \frac{1 + \cos E}{2} = \cos^2\frac{E}{2}\right)$$

$$\Rightarrow \sqrt{r}\cos\frac{\nu}{2} = \sqrt{a(1 - e)}\cos\frac{E}{2} \quad \text{(i)}$$

$$\sin^2\frac{\nu}{2} = \frac{1 - \cos\nu}{2}$$

$$\Rightarrow \sqrt{r}\sin\frac{\nu}{2} = \pm\sqrt{\frac{r(1 - \cos\nu)}{2}}$$

$$\Rightarrow \quad = \pm\sqrt{\frac{1}{2}a(1 + e)\frac{1 - \cos E}{2}}$$

$$\left(\Uparrow \frac{1 - \cos E}{2} = \sin^2\frac{E}{2}\right)$$

$$\Rightarrow \sqrt{r}\sin\frac{\nu}{2} = \sqrt{a(1 + e)}\sin\frac{E}{2} \quad \text{(ii)}$$

$$\sqrt{r}\sqrt{r_0}\cos\left(\frac{\nu\pm\nu_0}{2}\right)=\left[\sqrt{r}\cos\frac{\nu}{2}\right]\left[\sqrt{r_0}\cos\frac{\nu_0}{2}\right]\mp\left[\sqrt{r}\sin\frac{\nu}{2}\right]\left[\sqrt{r_0}\sin\frac{\nu_0}{2}\right]$$

$$\left(\Uparrow\cos\left(\frac{\nu\pm\nu_0}{2}\right)=\cos\left(\frac{\nu}{2}\right)\cos\left(\frac{\nu_0}{2}\right)\mp\sin\left(\frac{\nu}{2}\right)\sin\left(\frac{\nu_0}{2}\right)\right)$$

$$\left(\begin{array}{l}\Uparrow\sqrt{r}\cos\frac{\nu}{2}=\sqrt{a(1-e)}\cos\frac{E}{2},\ \sqrt{r_0}\cos\frac{\nu_0}{2}=\sqrt{a(1-e)}\cos\frac{E_0}{2}\\ \sqrt{r}\sin\frac{\nu}{2}=\sqrt{a(1+e)}\sin\frac{E}{2},\ \sqrt{r_0}\sin\frac{\nu_0}{2}=\sqrt{a(1+e)}\sin\frac{E_0}{2}\end{array}\right)$$

$$=a(1-e)\cos\frac{E}{2}\cos\frac{E_0}{2}\mp a(1+e)\sin\frac{E}{2}\sin\frac{E_0}{2}$$

$$=a\left(\cos\frac{E}{2}\cos\frac{E_0}{2}\mp\sin\frac{E}{2}\sin\frac{E_0}{2}\right)-ae\left(\cos\frac{E}{2}\cos\frac{E_0}{2}\pm\sin\frac{E}{2}\sin\frac{E_0}{2}\right)$$

$$=a\cos\left(\frac{E}{2}\pm\frac{E_0}{2}\right)-ae\cos\left(\frac{E}{2}\mp\frac{E_0}{2}\right)\qquad(iii)$$

$$\sqrt{rr_0}\left[cos\left(\frac{v\text{-}v_0}{2}\right)+ecos\left(\frac{v+v_0}{2}\right)\right]=acos\left(\frac{E\text{-}E_0}{2}\right)\text{-}aecos\left(\frac{E+E_0}{2}\right)+aecos\left(\frac{E+E_0}{2}\right)\text{-}ae^2cos\left(\frac{E\text{-}E_0}{2}\right)$$

$$=acos\left(\frac{E\text{-}E_0}{2}\right)\text{-}ae^2cos\left(\frac{E\text{-}E_0}{2}\right)$$

$$=a(1\text{-}e^2)cos\left(\frac{E\text{-}E_0}{2}\right)$$

$$=pcos\left(\frac{E\text{-}E_0}{2}\right)$$

$$\Rightarrow e\cos\left(\frac{\nu+\nu_0}{2}\right)=\frac{p}{\sqrt{r_0r}}\cos\left(\frac{E-E_0}{2}\right)-\cos\left(\frac{\nu-\nu_0}{2}\right)\qquad(3)$$

Substituting (3) into (2) results in

$$p\left(\frac{1}{r_0}+\frac{1}{r}\right)=2+\frac{2p}{\sqrt{r_0r}}\cos\left(\frac{E-E_0}{2}\right)\cos\left(\frac{\nu-\nu_0}{2}\right)-2\cos^2\left(\frac{\nu-\nu_0}{2}\right)$$

$$\Rightarrow p\left[\frac{r_0+r}{r_0r}-\frac{2}{\sqrt{r_0r}}\cos\left(\frac{E-E_0}{2}\right)\cos\left(\frac{\nu-\nu_0}{2}\right)\right]=2-2\cos^2\left(\frac{\nu-\nu_0}{2}\right)$$

$$\Rightarrow p\left[\frac{r_0+r}{r_0r}-\frac{2}{\sqrt{r_0r}}\cos\left(\frac{E-E_0}{2}\right)\cos\left(\frac{\nu-\nu_0}{2}\right)\right]=2\sin^2\left(\frac{\nu-\nu_0}{2}\right)\qquad(4)$$

Rearranging (1) for $p$ yields

$$p = \frac{y^2 r^2 r_0^2 \sin^2(\nu - \nu_0)}{\mu \Delta t^2} \tag{5}$$

Substituting (5) into (4) yields

$$\frac{y^2 r^2 r_0^2 \sin^2(\nu - \nu_0)}{\mu \Delta t^2}\left[\frac{r_0 + r}{r_0 r} - \frac{2}{\sqrt{r_0 r}}\cos\left(\frac{E - E_0}{2}\right)\cos\left(\frac{\nu - \nu_0}{2}\right)\right] = 2\sin^2\left(\frac{\nu - \nu_0}{2}\right)$$

$$\Rightarrow y^2 = \frac{2\mu \Delta t^2 \sin^2\left(\frac{\nu - \nu_0}{2}\right)}{r r_0 \sin^2(\nu - \nu_0)\left[r_0 + r - 2\sqrt{r_0 r}\cos\left(\frac{E - E_0}{2}\right)\cos\left(\frac{\nu - \nu_0}{2}\right)\right]}$$

$$\left(\Uparrow \sin(\nu - \nu_0) = 2\sin\left(\frac{\nu - \nu_0}{2}\right)\cos\left(\frac{\nu - \nu_0}{2}\right)\right)$$

$$\Rightarrow y^2 = \frac{\mu \Delta t^2 \sec^2\left(\frac{\nu - \nu_0}{2}\right)}{2 r r_0\left[r_0 + r - 2\sqrt{r_0 r}\cos\left(\frac{E - E_0}{2}\right)\cos\left(\frac{\nu - \nu_0}{2}\right)\right]} \tag{6}$$

If we define $l$, $m$, $x$ such that

$$l \equiv \frac{r_0 + r}{4\sqrt{r_0 r}\cos\left(\frac{\nu - \nu_0}{2}\right)} - \frac{1}{2}$$

$$m \equiv \frac{\mu \Delta t^2}{\left[2\sqrt{r_0 r}\cos\left(\frac{\nu - \nu_0}{2}\right)\right]^3}$$

$$x \equiv \frac{1}{2}\left[1 - \cos\left(\frac{E - E_0}{2}\right)\right] = \sin^2\left(\frac{E - E_0}{4}\right) \qquad \Leftarrow \sin^2 z = \frac{1 - \cos 2z}{2}$$

then, (6) becomes **the First Equation of Gauss**:

$$y^2 = \frac{m}{l + x} \tag{7}$$

which provides a relationship between the differences in eccentric anomalies ($E - E_0$) and the ratio of the sector to triangle areas.

$(m, l)$ are functions of $(\vec{r}_0, \vec{r})$, and thus is known, but $x$ needs to be determined.

Kepler's equation

$$\begin{cases} \dfrac{\sqrt{\mu}}{a^{3/2}}(t-T) = E - e\sin E \\ \dfrac{\sqrt{\mu}}{a^{3/2}}(t_0-T) = E_0 - e\sin E_0 \end{cases}$$

$$\Rightarrow \frac{\sqrt{\mu}}{a^{3/2}}(t-t_0)=E-E_0-e(\sin E-\sin E_0)$$

$$\left(\Uparrow\ \sin z-\sin z_0=2\sin\left(\frac{z-z_0}{2}\right)\cos\left(\frac{z+z_0}{2}\right)\right)$$

$$=E-E_0-2e\sin\left(\frac{E-E_0}{2}\right)\cos\left(\frac{E+E_0}{2}\right)$$

$$\left(\Uparrow\ e\cos\left(\frac{E+E_0}{2}\right)=\cos\left(\frac{E-E_0}{2}\right)-\frac{\sqrt{r_0 r}}{a}\cos\left(\frac{\nu-\nu_0}{2}\right)\Leftarrow (iii)\right)$$

$$\Rightarrow \frac{\sqrt{\mu}}{a^{3/2}}\Delta t=\left(E-E_0\right)-2\sin\left(\frac{E-E_0}{2}\right)\cos\left(\frac{E-E_0}{2}\right)+2\frac{\sqrt{r_0 r}}{a}\sin\left(\frac{E-E_0}{2}\right)\cos\left(\frac{\nu-\nu_0}{2}\right)$$

$$\left(\Uparrow\ \sin(z-z_0)=2\sin\left(\frac{z-z_0}{2}\right)\cos\left(\frac{z-z_0}{2}\right)\right)$$

$$=E-E_0-\sin(E-E_0)+2\frac{\sqrt{r_0 r}}{a}\sin\left(\frac{E-E_0}{2}\right)\cos\left(\frac{\nu-\nu_0}{2}\right)$$

$$\sin\left(\frac{E-E_0}{2}\right)=\sin\frac{E}{2}\cos\frac{E_0}{2}-\cos\frac{E}{2}\sin\frac{E_0}{2}$$

$$\text{from}\,(i)\ and\ (ii)\rightarrow\left(\begin{array}{ll} \sin\dfrac{E}{2}=\dfrac{\sqrt{r}\sin\dfrac{\nu}{2}}{\sqrt{a(1+e)}}, & \cos\dfrac{E_0}{2}=\dfrac{\sqrt{r_0}\cos\dfrac{\nu_0}{2}}{\sqrt{a(1-e)}} \\ \cos\dfrac{E}{2}=\dfrac{\sqrt{r}\cos\dfrac{\nu}{2}}{\sqrt{a(1-e)}}, & \sin\dfrac{E_0}{2}=\dfrac{\sqrt{r_0}\sin\dfrac{\nu_0}{2}}{\sqrt{a(1+e)}}, \end{array}\right)$$

$$=\frac{\sqrt{r_0 r}}{\sqrt{a^2(1-e^2)}}\left(\sin\frac{\nu}{2}\cos\frac{\nu_0}{2}-\cos\frac{\nu}{2}\sin\frac{\nu_0}{2}\right)$$

$$=\frac{\sqrt{r_0 r}}{\sqrt{ap}}\sin\left(\frac{\nu-\nu_0}{2}\right) \qquad (iv)$$

$$\Rightarrow \Delta t=\frac{a^{3/2}}{\sqrt{\mu}}[E-E_0-\sin(E-E_0)]+2\frac{a^{3/2}}{\sqrt{\mu}}\frac{\sqrt{r_0 r}}{a}\frac{\sqrt{r_0 r}}{\sqrt{ap}}\sin\left(\frac{\nu-\nu_0}{2}\right)\cos\left(\frac{\nu-\nu_0}{2}\right)$$

$$=\frac{a^{3/2}}{\sqrt{\mu}}[E-E_0-\sin(E-E_0)]+\frac{r_0 r\sin(\nu-\nu_0)}{\sqrt{\mu p}} \qquad (8)$$

Rearranging (1) leads to

$$\frac{1}{y} = \frac{r_0 r \sin(\nu - \nu_0)}{\sqrt{\mu p}\,\Delta t}$$

$$= \frac{r_0 r}{\sqrt{\mu p}\,\Delta t}\left[\Delta t - \frac{a^{3/2}}{\sqrt{\mu}}\left[E - E_0 - \sin(E - E_0)\right]\right]\frac{\sqrt{\mu p}}{r_0 r}$$

$$= 1 - \frac{a^{3/2}}{\Delta t\sqrt{\mu}}[E - E_0 - \sin(E - E_0)]$$

$$\Rightarrow 1 - \frac{1}{y} = \frac{a^{3/2}}{\Delta t\sqrt{\mu}}[E - E_0 - \sin(E - E_0)] \tag{9}$$

$$\sin(\nu - \nu_0) = 2\sin\left(\frac{\nu - \nu_0}{2}\right)\cos\left(\frac{\nu - \nu_0}{2}\right)$$

$$\left(\Uparrow \sin\left(\frac{\nu - \nu_0}{2}\right) = \frac{\sqrt{ap}}{\sqrt{r_0 r}}\sin\left(\frac{E - E_0}{2}\right) \Leftarrow (iv)\right)$$

$$\Rightarrow \sin(\nu - \nu_0) = 2\frac{\sqrt{ap}}{\sqrt{r_0 r}}\sin\left(\frac{E - E_0}{2}\right)\cos\left(\frac{\nu - \nu_0}{2}\right) \tag{10}$$

From (1),

$$y = \frac{\sqrt{\mu p}\,\Delta t}{r_0 r \sin(\nu - \nu_0)}$$

$$\Rightarrow y = \frac{\sqrt{\mu}\,\Delta t}{2\sqrt{a}\sqrt{r_0 r}\,\sin\left(\frac{E - E_0}{2}\right)\cos\left(\frac{\nu - \nu_0}{2}\right)} \tag{11}$$

Eliminating $a$ from (9) and (11),

$$(11)^3 \;\Rightarrow\; y^3 = \frac{\mu\sqrt{\mu}\,\Delta t^3}{a^{3/2}\left[2\sqrt{r_0 r}\,\sin\left(\frac{E - E_0}{2}\right)\cos\left(\frac{\nu - \nu_0}{2}\right)\right]^3}$$

$$\Rightarrow y^3\left(1 - \frac{1}{y}\right) = \frac{\mu\,\Delta t^2}{\left[2\sqrt{r_0 r}\,\cos\left(\frac{\nu - \nu_0}{2}\right)\right]^3}\;\frac{[E - E_0 - \sin(E - E_0)]}{\sin^3\left(\frac{E - E_0}{2}\right)} \tag{12}$$

$$\left( \Uparrow \; X \equiv \frac{[E - E_0 - \sin(E - E_0)]}{\sin^3 \left(\frac{E - E_0}{2}\right)} \right) \tag{13}$$

We obtain **the Second Equation of Gauss**:

$$y^2(y-1) = mX \tag{14}$$

So far, three basic equations have been obtained:

I. (1) the area ratio of sector to triangle
II. (7) the first equation of Gauss
III. (14) the second equation of Gauss

Recall that (*m,l*) are known from the given position vectors $(\vec{r}_0, \vec{r})$

However, (*x,X,y*) are unknown, and needs to be determined:

I. Substitute (7) into (14) leads to

$$\frac{m}{l+x}(y-1) = mX$$

$$y = 1 + X(l+x) \tag{15}$$

II. (7) can be rearranged as

$$x = \frac{m}{y^2} - l \tag{16}$$

III. Iterative procedure involving (15) and (16) enables us to determine *x*, with the initial guess for *y=1* (Note *y* is the area ratio)

Once *x* is known, the difference $(E - E_0)$ can be computed by the definition of *x*:

$$x \equiv \frac{1}{2}\left[1 - \cos\left(\frac{E - E_0}{2}\right)\right]$$

$$\Rightarrow \cos\left(\frac{E - E_0}{2}\right) = 1 - 2x \tag{17}$$

Since $\frac{E - E_0}{2} \in [0, \pi] \left(\Leftrightarrow (E - E_0) \in [0, 2\pi]\right)$ , $(E - E_0)$ is uniquely determined

IV. *X* can be determined from (13)
V. *y* can be computed from (15), and is substituted into (16) to update *x*
VI. Repeat I to V until $|(y_0 - y_i)| \le \varepsilon$ is satisfied

VII. Once $y$ is determined, the semi-major axis and semi-parameter can be computed from (11) and (5), respectively:

$$a = \frac{\mu \Delta t^2}{4y^2 r_0 r \sin^2(\frac{E-E_0}{2})\cos^2(\frac{\nu-\nu_0}{2})} \tag{18}$$

$$p = \frac{y^2 r^2 r_0^2 \sin^2(\nu-\nu_0)}{\mu \Delta t^2} \tag{19}$$

We can use the $f$ and $g$ functions to determine velocity vectors, (and ultimately 6 orbital elements):

$$\begin{cases} f = 1 - \frac{r}{p}\left[1-\cos(\nu-\nu_0)\right] \\ g = \frac{r r_0 \sin(\nu-\nu_0)}{\sqrt{\mu p}} \\ \dot{f} = \sqrt{\frac{1}{p}}\tan\left(\frac{\nu-\nu_0}{2}\right)\left[\frac{1-\cos(\nu-\nu_0)}{p} - \frac{1}{r} - \frac{1}{r_0}\right] \\ \dot{g} = 1 - \frac{r_0}{p}\left[1-\cos(\nu-\nu_0)\right] \end{cases}$$

$$\Rightarrow \begin{cases} \vec{r} = f\,\vec{r}_0 + g\,\vec{v}_0 \\ \vec{v} = \dot{f}\,\vec{r}_0 + \dot{g}\,\vec{v}_0 \end{cases}$$

$$\Rightarrow \begin{cases} \vec{v}_0 = \frac{\vec{r} - f\,\vec{r}_0}{g} \\ \vec{v} = \dot{f}\,\vec{r}_0 + \dot{g}\left(\frac{\vec{r} - f\,\vec{r}_0}{g}\right) \\ \quad = \frac{\dot{g}\vec{r}}{g} - \frac{1}{g}(\dot{g}f - g\dot{f})\vec{r}_0 \quad \Leftarrow \dot{g}f - g\dot{f} = 1 \\ \quad = \frac{\dot{g}\vec{r} - \vec{r}_0}{g} \end{cases}$$

- Find $\vec{v}_0, \vec{v}$ using $(\vec{r}_0, \vec{r}, \Delta t)$.
- Determine 6 orbital elements for $(\vec{r}_0,\ \vec{v}_0)$ *or* $(\vec{r},\ \vec{v})$

## 7.7 Application: Targeting Problem

- Solution to Lambert Problem allows us to study some useful situations
  - Initial Orbit Determination
    - Yield Velocity Vectors from a set of observed Position Vectors
  - Can be viewed as a Transfer between Two Position Vectors in One Orbit
    - Can be used to plan a maneuver of a satellite between two separate positions
  - Orbit Transfer Problem
    - Position Vectors represent places in Two Separate Orbits
    - Applicable to complex maneuvers of Orbital Maneuvering in Chapter 6
    - Also a technique for Initial Orbit Determination because we try to establish an Orbit for the Transfer.
    - Intercept Maneuver: usually employ a Single Impulse
    - Targeting or Rendezvous Maneuver: usually use Two Burns

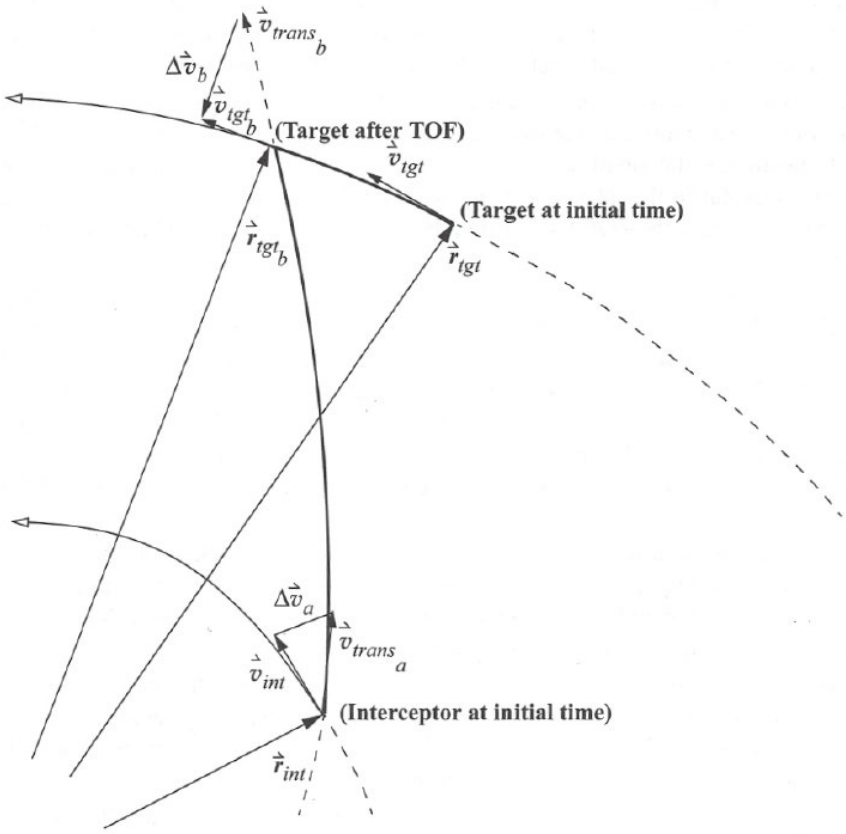



**Figure 7-14. Geometry for Intercept Maneuvers.** For the targeting maneuver to take place, we *must* account for the target's motion during the transfer.

- Lambert Problem for analyzing the Transfer Between Orbits and the necessary Timing
  - Tackle many complex maneuvers for which the analysis in Chapter 6 is difficult to apply.
    - Fig. 7-14 shows the overall geometry
    - Know the initial position and velocity vectors of Interceptor and Target
    - Also know either the desired time of flight (TOF) to complete the maneuver or the maximum change in velocity of the interceptor
    - Methods of solving the targeting problem differ depending on what else is known

  - First Step is to consider the motion of Target, which must be propagated from its initial position to where it will be after the TOF
  - Second Phase deals with transferring the interceptor from its initial position to that of the target satellite after it has moved through TOF
    - Recall that this tries to find a transfer between two known position vectors.

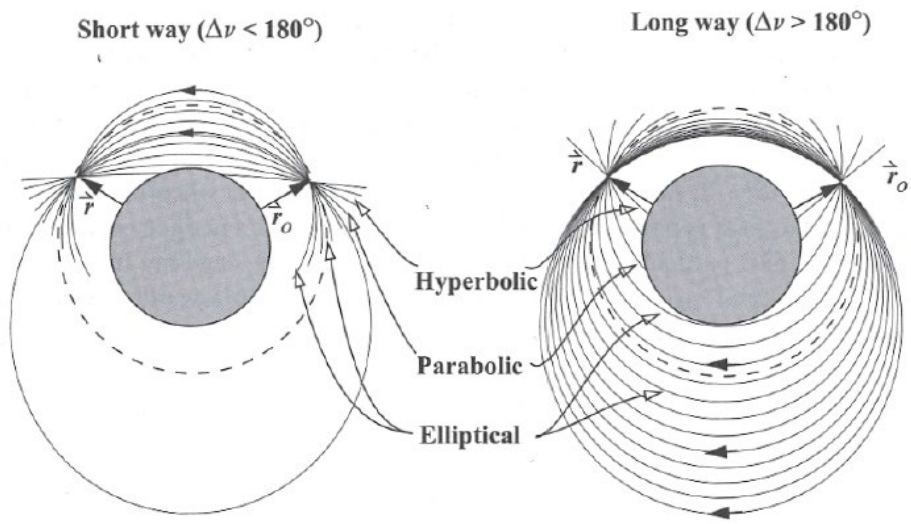



**Figure 7-15. Varying Time of Flight for Intercept.** As the time of flight increases for the transfer to the target (short way left, long way right), the transfer orbit becomes less eccentric until it reaches a minimum, shown as a dashed line. The eccentricity then begins to increase after this transfer. There is always a minimum eccentricity transfer, and a minimum change in velocity transfer. However, they generally don't occur at the same time. Finally, I've shown the initial and final positions with the same magnitude. This is not an additional requirement.

  - Short Way to the target is almost always used because of time constraints
  - Long Way can save more fuel in many circumstances
    - Should consider when trying to determine the smallest change in velocity
  - As we increase the TOF for transfer, the transfer trajectory can change through each type of conic section: Figure 7-15
  - Before trying to work through various results, we must check all solutions to ensure the transfer trajectory does not intersect the Earth: Fig 7-16

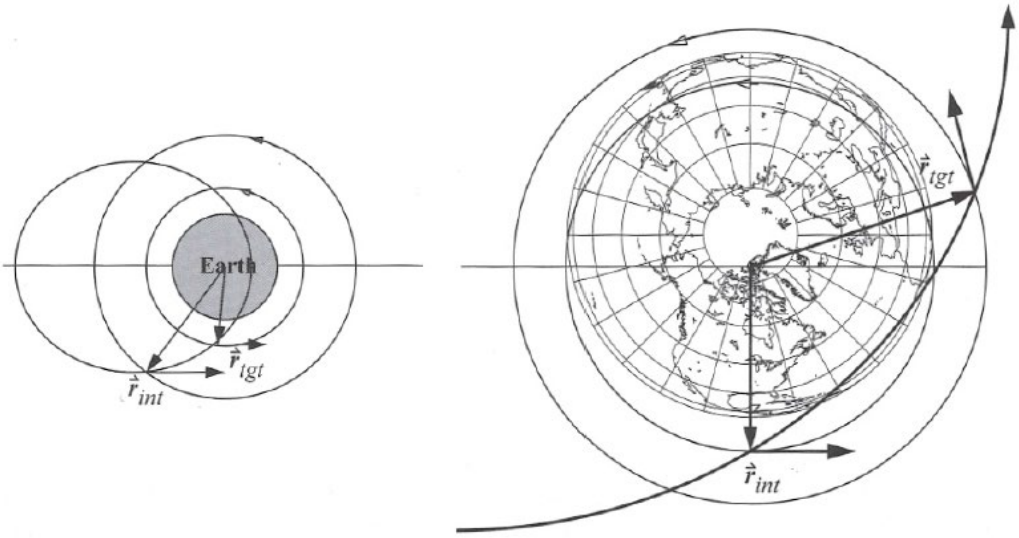



**Figure 7-16. Geometry of Orbits that Can Hit the Earth.** The orbit transfer on the left is valid because the transfer orbit intersects the Earth after the transfer. The transfer on the right would be unacceptable.

***ALGORITHM 59:*** ***HIT EARTH ALGORITHM 60:***

$(\vec{r}_{int}, \vec{r}_{tgt}, \vec{v}_{trans_a}, \vec{v}_{trans_b}, a \Rightarrow r_p,$ collision )

IF $((\vec{r}_{int} \cdot \vec{v}_{trans_a}) < 0.0)$ and $((\vec{r}_{tgt} \cdot \vec{v}_{trans_b}) > 0.0)$

$$\xi = \frac{v_{trans_a}^2}{2} - \frac{\mu}{r_{int}} = -\frac{\mu}{2a}$$

$$h_t = \left|\vec{r}_{int} \times \vec{v}_{trans_a}\right|$$

$$p = \frac{h_t^2}{\mu}$$

$$e = \sqrt{\frac{a-p}{a}}$$

$$r_p = a(1-e)$$

IF $r_p \le R_\oplus$ (1.0 ER, 6378.136 km)

Collision

ELSE

No Collision

- Use of Lambert Problem for Targeting using only Two-Body dynamics

Algorithm 61: TARGET $\left(\vec{r}_{int}, \vec{r}_{tgt}, \vec{v}_{int}, \vec{v}_{tgt}, \Delta t \Rightarrow \Delta\vec{v}_a, \Delta\vec{v}_b\right)$

I. Propagate Target through the desired TOF

PKepler $\left(\vec{r}_{tgt}, \vec{v}_{tgt}, \Delta t \Rightarrow \vec{r}_{tgt,b}, \vec{v}_{tgt,b}\right)$

II. Determine Transfer Orbit using Lambert Problem Approach

Lambert-Gauss $(\vec{r}_{\text{int}}, \vec{r}_{tgt,b}, \Delta t \Rightarrow \vec{v}_{trans,a}, \vec{v}_{trans,b})$

III. $$\begin{aligned} \Delta\vec{v}_a &= \vec{v}_{trans,a} - \vec{v}_{\text{int}} \\ \Delta\vec{v}_b &= \vec{v}_{tgt} - \vec{v}_{trans,b} \end{aligned}$$

IV. Check for the Earth impact

Hit Earth $(\vec{r}_{\text{int}}, \vec{v}_{trans,a}, \vec{r}_{tgt,b}, \vec{v}_{trans,b}, a \Rightarrow r_p, collision)$

To gain insight into the two steps for solving this problem, first examine the solution and some characteristics of the Lambert problem. Recall Fig. 7-15, which showed two satellite position vectors and the range of trajectories between them. These represent solutions of Lambert's problem using the universal-variable routine for various TOFs. To fully examine the range of potential solutions, it's useful to look at an extended problem. We'll vary time and examine two cases: (1) a fixed target and interceptor, and (2) a moving target and interceptor. The fixed case isn't realistic, but it lays a foundation to compare with moving methods. It also provides a reference to the simplified techniques in Chap. 6. Figure 7-17 shows the change in velocity between two specified position vectors for the short- and long-way trajectories. The initial conditions are:

INITIAL CONDITIONS FOR FIXED-TARGET PROBLEM

INTERCEPTOR $\vec{r}_{IJK} = -6518.1083\ \hat{I} - 2403.8479\ \hat{J} - 22.1722\ \hat{K}$ km

$\vec{v}_{IJK} = 2.604\,057\ \hat{I} - 7.105\,717\ \hat{J} - 0.263\,218\ \hat{K}$ km/s

TARGET $\vec{r}_{IJK} = 6697.4756\ \hat{I} + 1794.5832\ \hat{J} + 0.0\ \hat{K}$ km

$\vec{v}_{IJK} = -1.962\,373\ \hat{I} + 7.323\,674\ \hat{J} + 0.0\ \hat{K}$ km/s

I've kept the scale for each figure constant to give you a relative sense of equality. The long-way total $\Delta v$ is usually greater than 30 km/s and doesn't appear on the figures. The initial positions of the interceptor and the target are about 175° apart.

As time increases, the velocity decreases to a local minimum. The minimum change in velocity for Fig. 7-17 occurs at 46.68 minutes on the short-way trajectory; $\Delta v_a = 1.88$ m/s, and $\Delta v_b = 265$ m/s. The step-size was adjusted for the interval around 46 minutes to

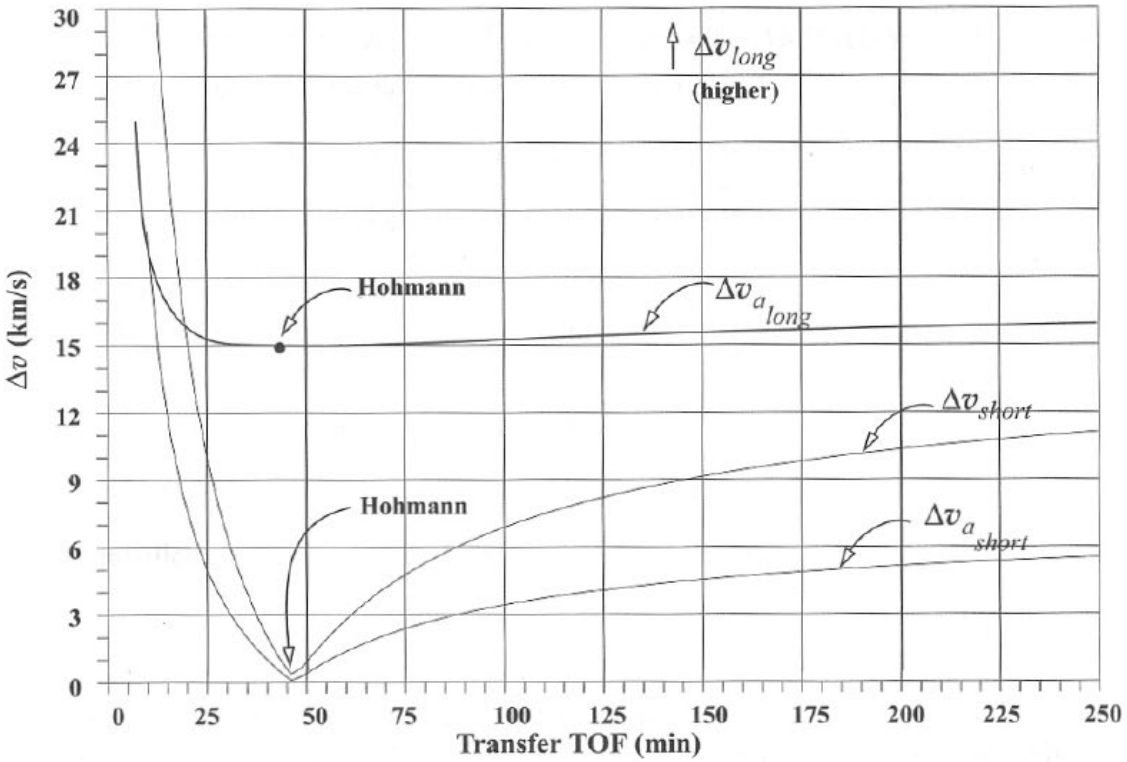



**Figure 7-17. Short- and Long-Way $\Delta v$ for Varying TOF, Target Fixed.** As the time of flight increases, the change in velocity reaches a minimum at the point where the transfer mimics a Hohmann transfer. The first, and total change in velocities ($\Delta v_a$, $\Delta v$) are shown for both short and long transfers.

determine this specific value. This same minimum velocity may also be calculated analytically using Escobal's method. Refer to Escobal ([1968] 1979:81–90).

Because the target is about 175° ahead of the interceptor and the target position is fixed, notice how the short way (Fig. 7-17) produces a dramatically lower change in velocity, whereas the long way requires a relatively high $\Delta v$ for all transfer times. It turns out the short-way trajectory at 46.68 minutes is similar to a Hohmann transfer. The values for the combined plane change (2°) in this Hohmann-like transfer are about $\Delta v_a$ = 2.98 m/s and $\Delta v_b$ = 262 m/s. The long way to the target never achieves this value because it always has to reverse its velocity and thus has much larger values. In general, to get a minimum change in velocity, you want the transfer velocity to be in the *same* direction as your initial velocity.